

Deep Reinforcement Learning

Exploration

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Content

Content

- Recap
 - Exploration-exploitation dilemma
 - Multi-armed bandits
- Advanced exploration methods for tabular approaches
 - Upper confidence bounds
 - Thompson sampling
 - Information gain
- Exploration in continuous spaces
 - The counting problem
 - Intrinsic motivation
 - Curiosity-driven exploration

Where are we?

Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6	Brief introduction to deep learning	The basics for what comes next
7	Value function approximation	Value estimation with function approximation
8	Deep Q-learning	Q-learning with neural networks
9	Policy gradients	Direct optimization of the policy
10	Actor-critic algorithms	Improved policy gradients via value functions
11	Advanced algorithms (Part I): From policy gradient to PPO	The PG route to modern RL algorithms
12	Advanced algorithms (Part II): From Q -learning to Soft Actor-Critic	The AC route to modern RL algorithms
13	Exploration	How to explore in complex scenarios
	Model-Based Control	
	Advanced Topics	

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Lecture contents

Recap

Exploration-exploitation dilemma

Multi-armed bandits

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Random network distillation

References